



WHAT NUCLEI ARE LOCATED IN THE BRAIN STEM?

The brain stem contains 10 of the 12 pairs of cranial nuclei. They provide motor and sensory function to the tongue, larynx, facial muscles, and more.

THE BRAIN HAS
MORE THAN 30 SETS
OF NUCLEI, MOSTLY
PAIRED LEFT AND RIGHT



REGIONS OF THE BASAL GANGLIA

NUCLEUS	FUNCTION
Caudate nucleus	A motor processing center that involves procedural learning of movement patterns and conscious inhibition of reflex actions.
Putamen	A motor control center, associated with complex learned procedures such as driving, typing, or playing a musical instrument.
Globus pallidus	A voluntary motor control center that manages movements at a subconscious level. When damaged, it can create involuntary tremors.
Subthalamic nucleus	Although its precise function is not clear, this structure is thought to be linked to selecting a specific movement and inhibiting any competing options.
Substantia nigra	Plays a role in reward and movement. Symptoms of Parkinson's disease (see p.201) are associated with the death of dopamine neurons found here.
Amygdala	May play a part in integrating activity between basal ganglia and limbic system, thereby considered by some to be part of the basal ganglia.

Action selection

The basal ganglia have an important role in filtering out the noise of competing commands coming from the cortex and elsewhere in the forebrain. This process is called action selection, and it occurs entirely subconsciously through a series of pathways through the basal ganglia. Generally, these pathways block or inhibit a specific action by having the thalamus loop the signal back to the start point. However, when the pathway is silent, the action goes ahead.

Basal ganglia loops

The route of the pathway depends on the source of the inputs from the cortex or elsewhere in the forebrain. There are three main pathways, and each one is able to inhibit or select an action. The motor loop connects to the main movement control center, the prefrontal loop carries input from executive regions of the brain, while the limbic loop is governed by emotional stimuli.

